NLP – Text Generation

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NLP Techniques

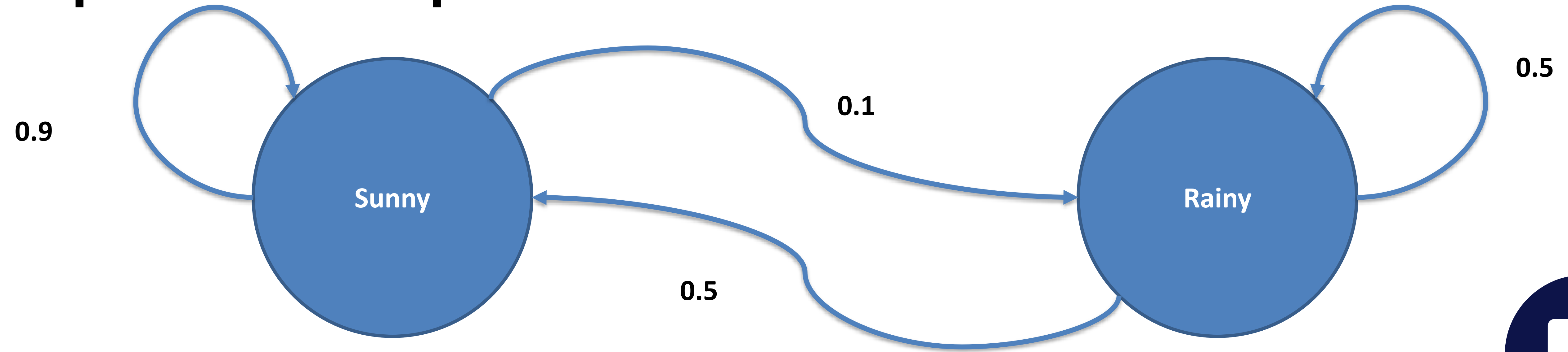
- Sentiment Analysis
- Topic Modeling
- Text Generation

Text Generation

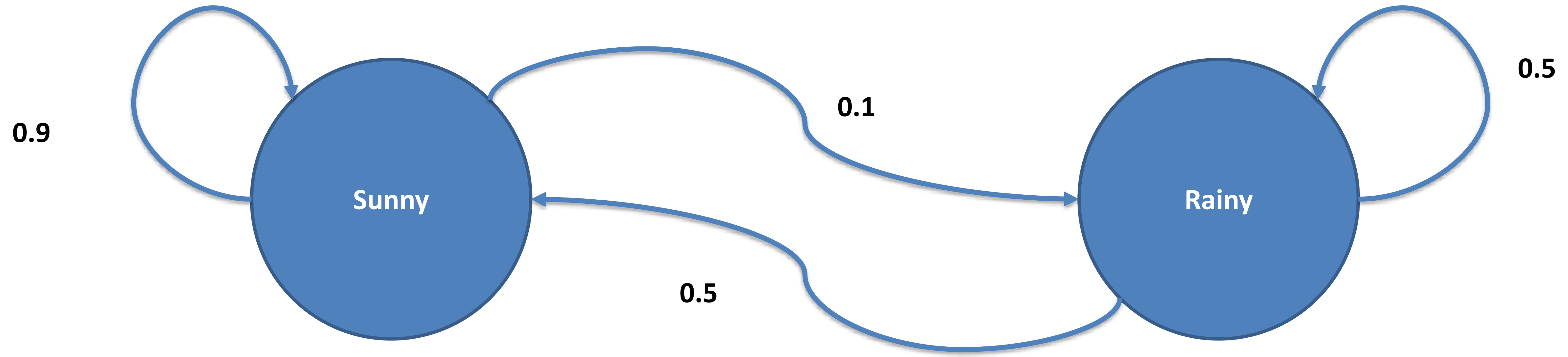
- Input: A corpus. We want to preserve the order of the text, including punctuation!
- Python: Using Markov Chains (based on the Markov Decision Process we've learned on deep learning) to do this and implementing Python's built-in functions
- Output: Our goal is to generate new movie transcripts in the style of each movie. This is more of a fun side project, but it's a cool way to look at the data and it has applications outside of text analysis

Markov Chains

- Markov chains are a way of representing **how systems change over time**
- The main concept behind Markov Chains are that they are memoryless, meaning that **the next state of a process only depends on the previous state**



Markov Chains



- The lines represent how they are connected to each other
- Because Markov Chains are based on probability statistics, the numbers represent the chances of how correlated two values may be, even if its compared to itself

Markov Chains

- Can you imagine how this would work for text generation?
- What would the circles represent?
- What would the arrows represent?

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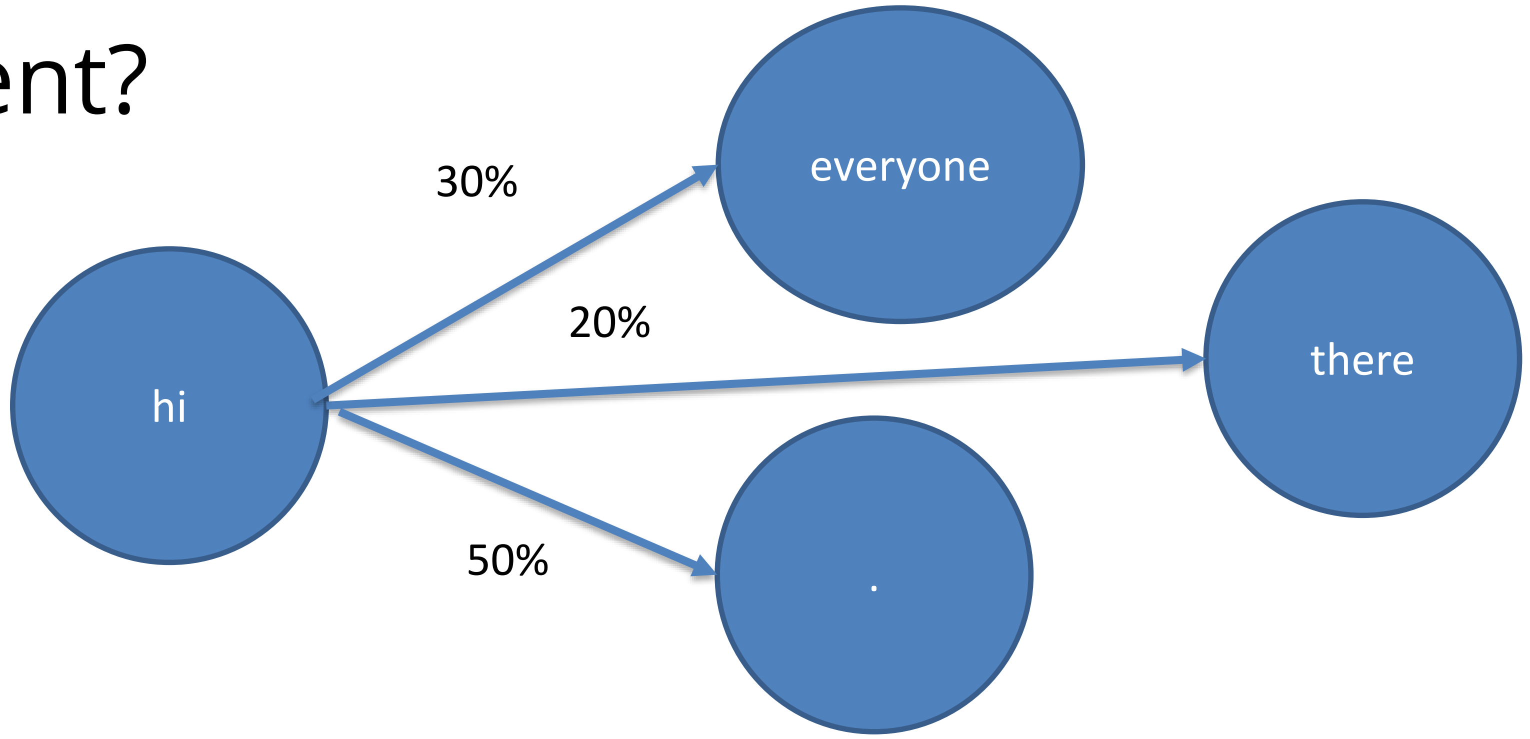

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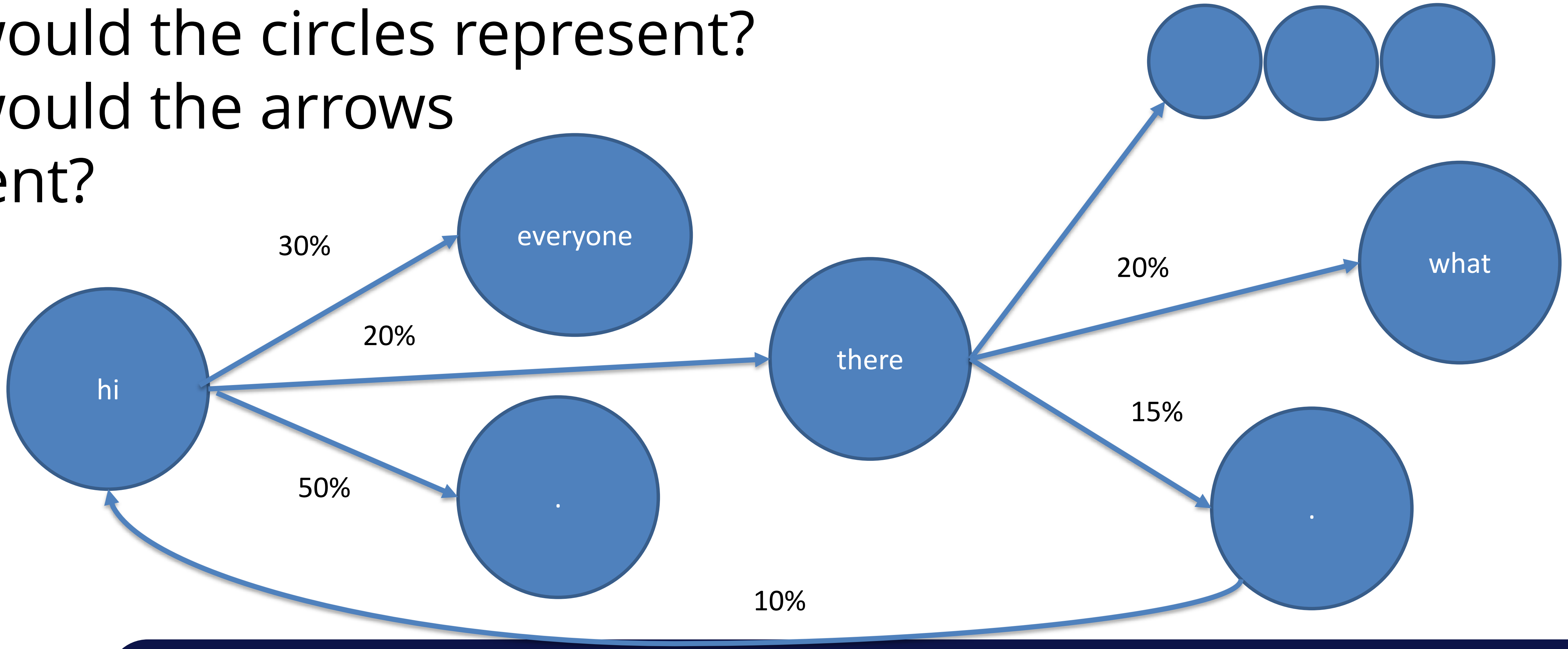

1. Question | 2. Data | 3. Perform EDA | 4. Techniques | 5. Insight

Markov Chains

This can be represented as a dictionary

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{‘hi’:[‘everyone’, ‘there’, ‘.’],
 ‘there’:[‘what’, ‘.’, …],
 ‘.’: [‘hi, ….]}
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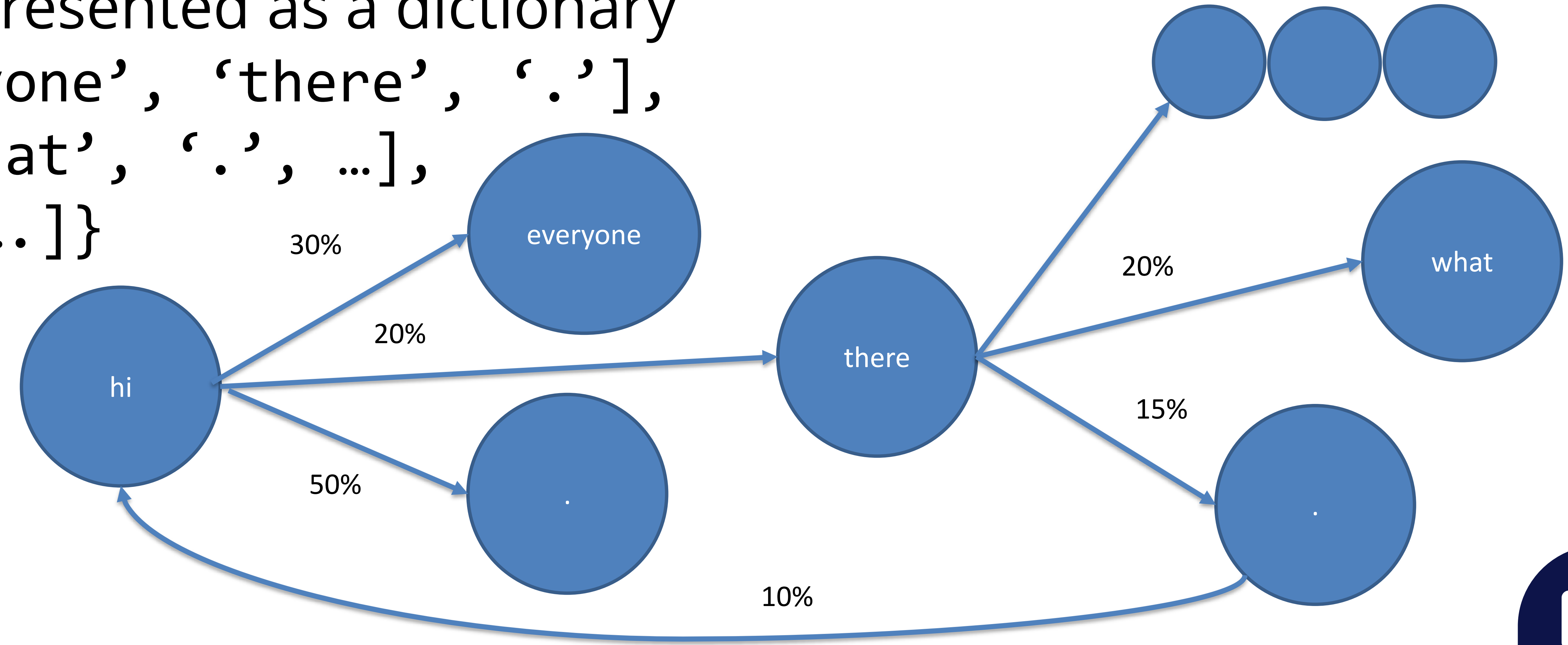
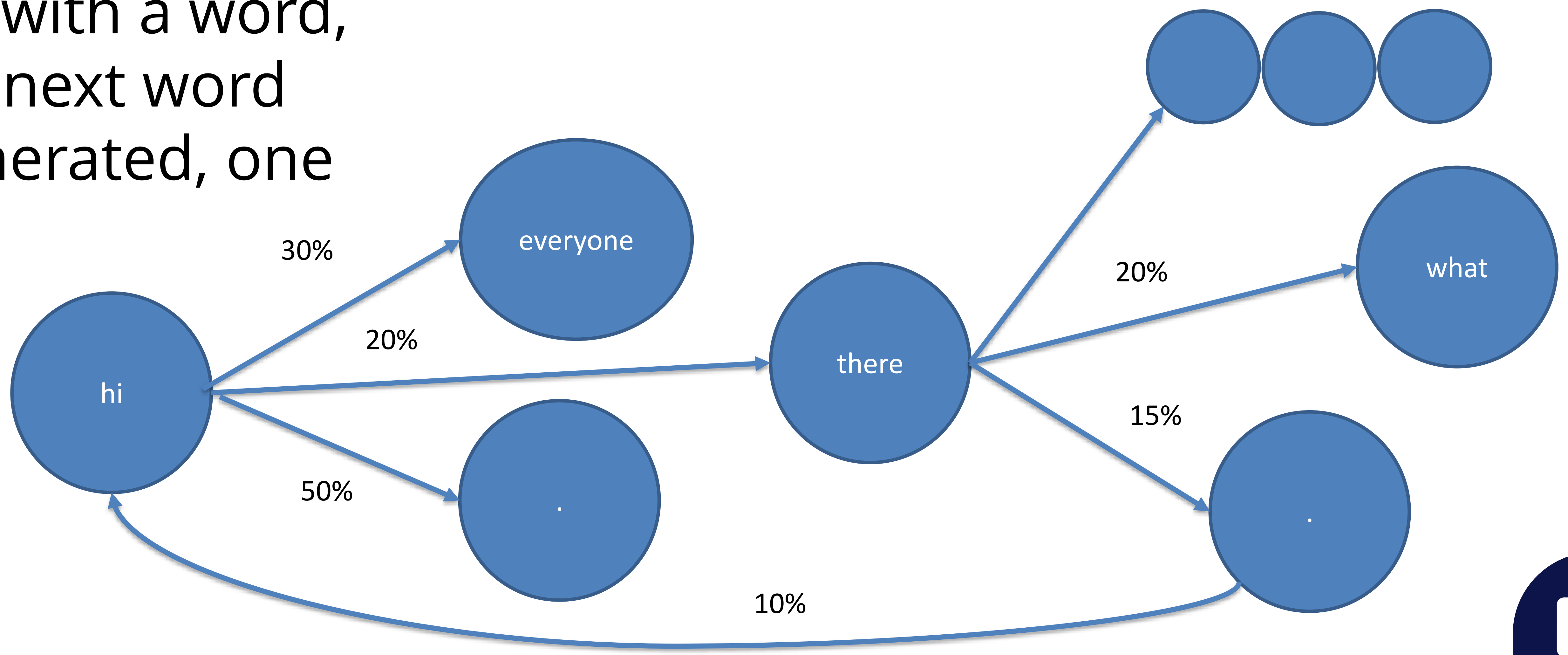


Markov Chains

You can start with a word, and have the next word randomly generated, one at a time



Markov Chains

- Create a dictionary for a corpus where the **keys are the current state** and the **values are the options for the next state**. Write a function to randomly generate next terms.
- Output: Movie transcripts in the style of each movie
- This is an oversimplified way of generating text. A much more complex technique would be to use deep learning and **Long Short-Term Memory (LSTM)**. This not only takes into account the previous word, but things like the words before the previous word, what words have already been generated etc.

Google Colab Code

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2. Data
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